lipo-boost — Highlights

Phase P9; all 5 recorded gates pass.

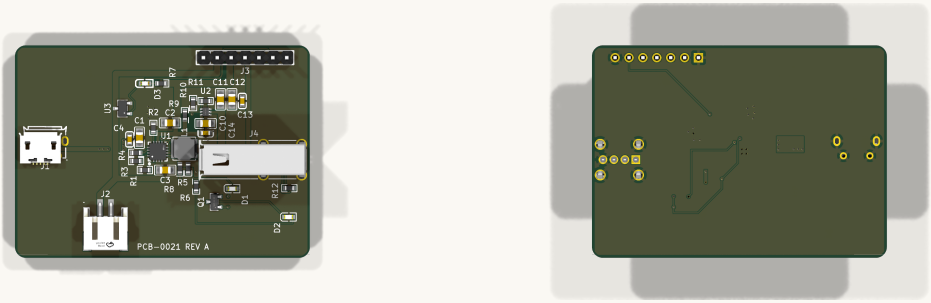
At a Glance

I want a little board that runs a 5 V USB gadget off a single-cell LiPo and charges the cell from USB.

- USB in (micro-B or C) charges a 1S LiPo on a JST-PH 2-pin connector, about 1 A max.
- Load sharing: when USB is plugged in, the 5 V output keeps running while the cell charges.
- 5 V / 1 A out on a USB-A socket, from a synchronous boost.
- An enable pin to turn the output off, plus a low-battery LED and charge / done LEDs.
- Small, cheap, JLCPCB-assembled, 2 layers is fine.

FIELD	VALUE
part number	PCB-0021-A
board	lipo-boost
phase	P9 (DFM)
size	53.9 x 38.8 mm
parts	33 placed parts, 24 BOM lines
gates	5 of 5 recorded gates pass

The Board

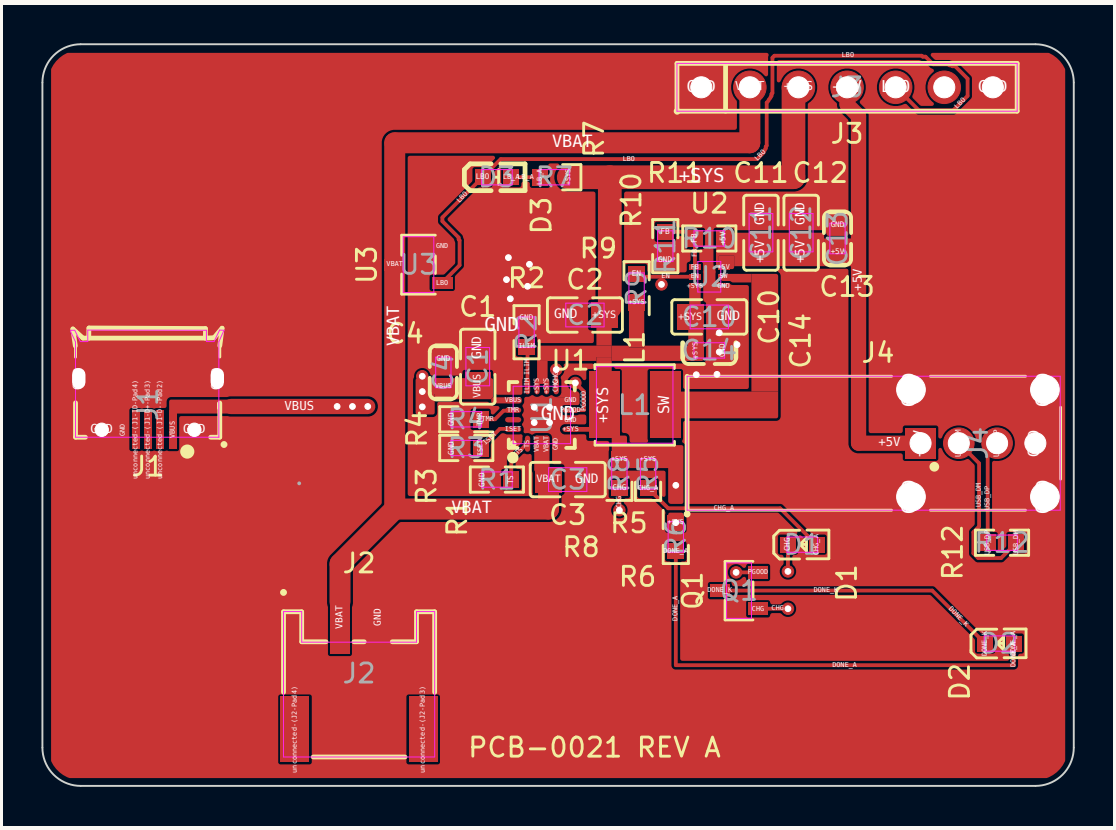


top and bottom, as rendered from the board file

Copper Layers

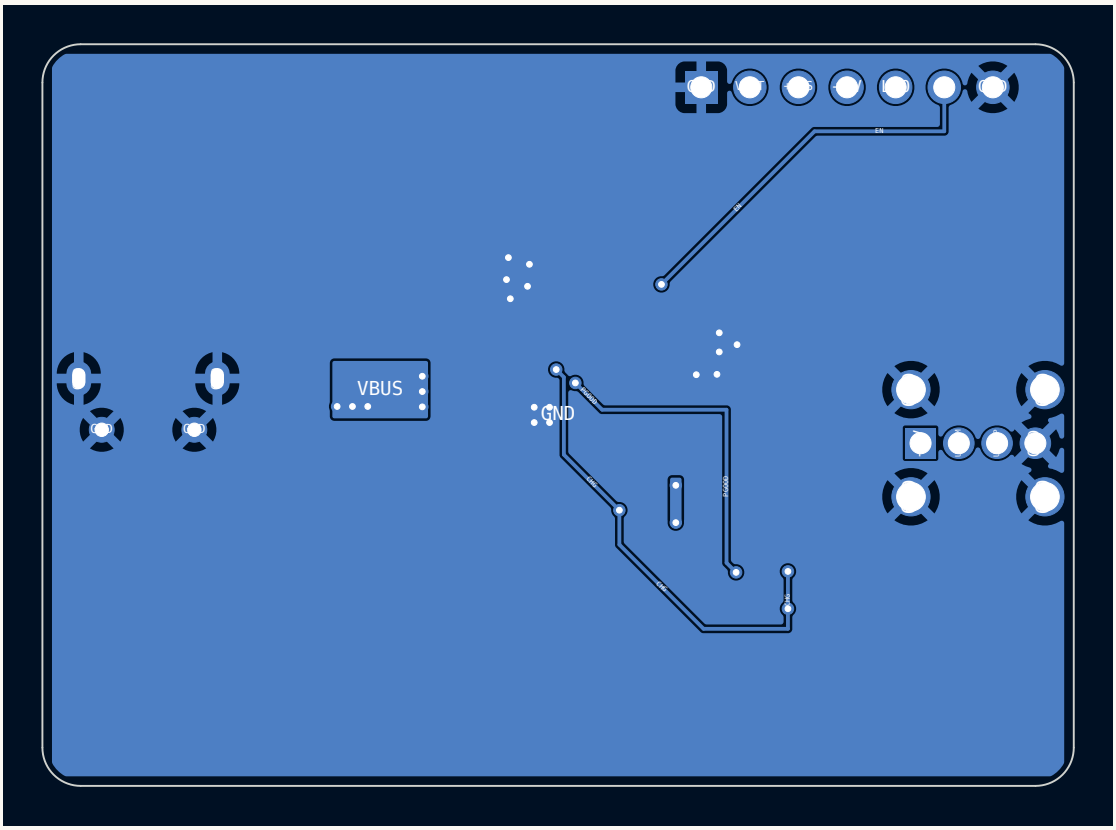
The outer copper layers as KiCad shows them, with every pad, track and zone labelled with its net (zoom in to read the small ones).

F.Cu



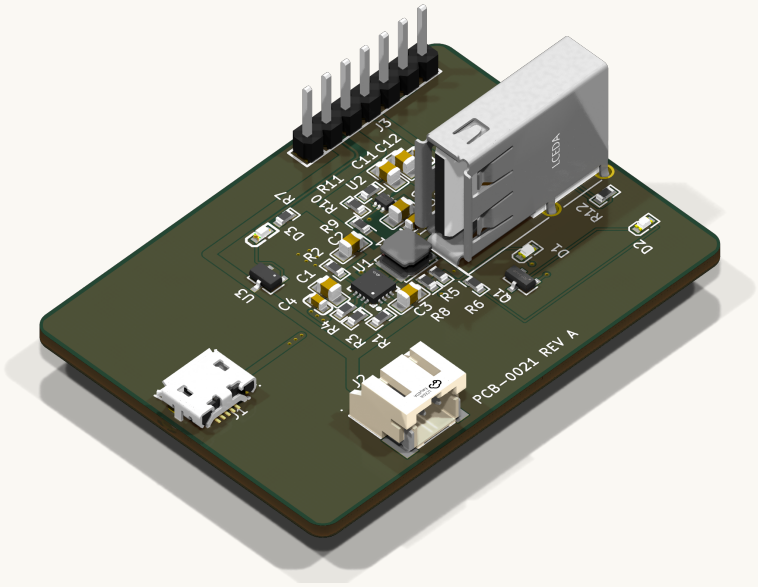
reports/layers/PCB-0021-A_lipo-boost_F_Cu.pdf

B.Cu



reports/layers/PCB-0021-A_lipo-boost_B_Cu.pdf

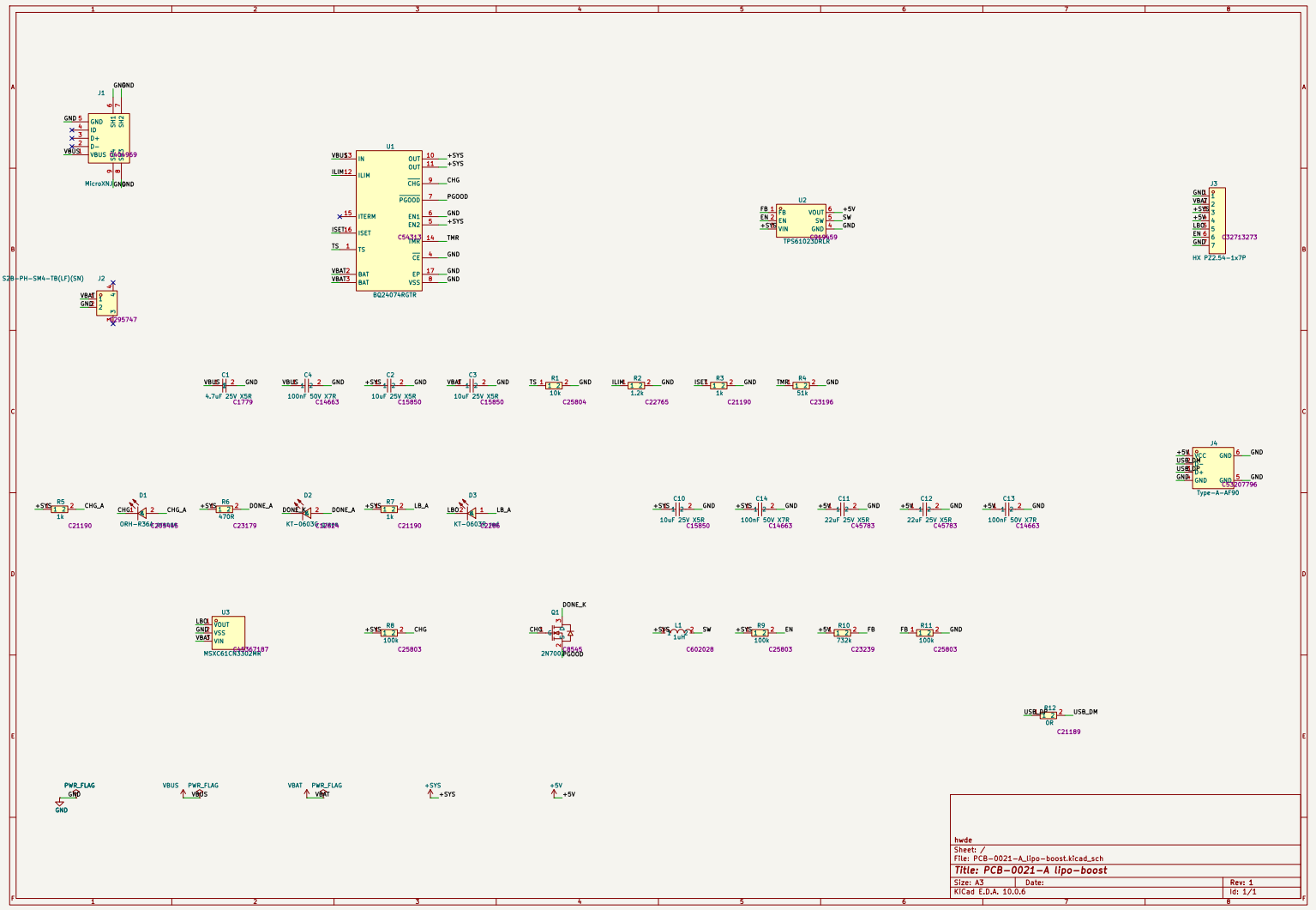
3D view



reports/layers/PCB-0021-A_lipo-boost_iso.png

What Is On It

REFS	PART	FOOTPRINT	LCSC
C1	4.7uF 25V X5R	C0805	C1779
C2,C3,C10	10uF 25V X5R	C0805	C15850
C4,C13,C14	100nF 50V X7R	C0603	C14663
C11,C12	22uF 25V X5R	C0805	C45783
D1	ORH-R36A orange	LED0603-RD	C205445
D2	KT-0603G green	LED0603-RD	C12624
D3	KT-0603R red	LED-SMD_L1.6-W0.8-R-RD	C2286
J1	MicroXNJ	MICRO-USB-SMD_MICROXNJ_1	C404969
J2	S2B-PH-SM4-TB(LF)(SN)	CONN-SMD_P2.00_S2B-PH-SM4-TB-LF-SN	C295747
J3	HX PZ2.54-1x7P	HDR-TH_7P-P2.54-V-M	C32713273
J4	Type-A-AF90	USB-A-TH_U-A-24DD-Y-13	C53207796
L1	1uH	IND-SMD_L4.0-W4.0	C602028
Q1	2N7002	SOT-23-3_L2.9-W1.3-P1.90-LS2.4-BR	C8545
R1	10k	R0603	C25804
R2	1.2k	R0603	C22765
R3,R5,R7	1k	R0603	C21190
R4	51k	R0603	C23196
R6	470R	R0603	C23179
R8,R9,R11	100k	R0603	C25803
R10	732k	R0603	C23239
R12	0R	R0603	C21189
U1	BQ24074RGTR	QFN-16_L3.0-W3.0-P0.50-TL-EP1.7	C54313
U2	TPS61023DRLR	SOT-563_L1.6-W1.2-P0.50-LS1.6-BR	C919459
U3	MSXC61CN3302MR	SOT-23-3_L2.9-W1.6-P1.90-LS2.8-BR	C45367187


J1
MicroXNJGNDND
U1
BQ24074RGTR
U2
TPS61023DRLR
J3
HX PZ2.54-1x7P
J2
SZH-PH-SM4-TB(LF)(SN)
J4
Type-A-AF90
D1
D2
D3
Q1
2N7002PDW
U3
M3XC61CR3302MR
C1
C4
C2
C3
R1
R2
R3
R4
R5
R6
R7
R8
R9
R10
R11
R12
C10
C14
C11
C12
C13
L1
PWR_FLAG
GND
VBUS
VBAT
+SYS
+5V
hwde
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File: PCB-0021-A_lipo-boost.kicad_sch
Title: PCB-0021-A lipo-boost
Size: A3
Date:
Rev: 1
KiCad E.D.A. 10.0.6
Id: 1/1

Key Decisions

PH	WHEN	DECISION	WHY
P2	10-01 05:51	Closed both P2 research tasks –accept-drafts (6/12 verified, 6 refuted drafts stay draft and do not inject)	budget; refutations are numeric corrections carried as the decision below, not wrong rules
P2	10-01 05:51	Design corrections from the second read: TPS61023 VREF 580/595/610 mV (PWM) for the FB divider tolerance; BQ24074 size ILIM/ISET with KILIM max 1720 (I = K/R); bq24074 VIN range 4.35-10.2 V (VOVP 10.2-10.8 V); TPS61023 RthJA 142.7 C/W -> ~0.3-0.55 W loss is a 43-78 C rise on 2L, so give U2 a GND pour and vias; ISET pads need no dedicated vias	second reader refutations, research/records
P3	10-01 06:00	PROVISIONAL: low-battery detector MSXC61CN3302MR C45367187 (3.3 V, not 3.2 V; releases ~3.45 V)	no stocked 3.2 V open-drain detector; 3.3 V warns a little earlier, harmless
P3	10-01 06:00	PROVISIONAL: RILIM 1.2k Basic (1.25-1.43 A) and RISET 1k Basic (~0.89 A) instead of 1.15k/910R Extended	two fewer extended-part fees; lower charge current is kinder to small cells; still about 1 A
P3	10-01 06:00	bq24074 pins: no SYSOFF/TD; ITERM open (10 %); EN2->SYS, EN1 and CE->GND (never float); TS 10k to GND; RTMR 51k (6.8 h)	part-sourcer read SLUS810N pin table; corrects architect's from-memory notes
P3	10-01 06:00	U1 BQ24074RGTR stock thin (417), same-die alternates RGTT C2861252 / RGTRG4	recheck before any order
P3	10-01 06:15	J4 swapped C18077685 -> C53207796 Type-A-AF90 (JXTCONN USB-A THT right angle, stock 180)	C18077685 not in EasyEDA API: no pullable symbol/footprint

How the Run Went

22 recorded decisions from P2 to P7, 2026-10-01T05:43 to 2026-10-01T08:01. The run went back to an earlier phase 0 times.

Versus Adafruit's PowerBoost 1000C

This board is a from-scratch recreation of Adafruit's **PowerBoost 1000C** (designed by Limor Fried / Ladyada for Adafruit Industries; github.com/adafruit/Adafruit-PowerBoost-1000C), published under **CC BY-SA 3.0**. It was designed from the brief and datasheets, with no Adafruit file copied, so it is not a derivative; the original is credited as the reference.

Same: a load-sharing charger (about 1 A into a 1S LiPo, output stays up on USB), a 5 V / 1 A synchronous boost, an EN pin, low-battery, charge and done LEDs, a USB-A output and a JST-PH battery connector, on two copper layers.

Different, and why:

- Charger BQ24074 (QFN-16, about 1,066 in stock, $1.23) rather than MCP73871 (QFN-20, about 112 in stock, $2.79). Its fixed 4.4 V system rail keeps the boost boosting, never passing USB through to the output.
- Boost TPS61023 (SOT-563, $0.27) rather than TPS61090 (QFN-16, $2.85). It has no low-battery comparator, so a separate 3.2 V detector (U3) drives the LBO output.
- Output about 4.95 V, inside the USB-A limit; the original is about 5.2 V by design. Inductor 1 uH, not 6.8 uH.
- Size 53.9 x 38.8 mm against about 36 x 23 mm for the original (its size is from the product page, not verified). Above the roughly 40 x 30 mm soft target in the requirements.
- All 33 parts are JLCPCB-assembled; every IC is a JLC Extended part. A fixed 10 k resistor stands in for the cell thermistor.

Open items for the owner (provisional waivers, agent-proposed 2026-10-01; verify passes only with them):

- U1 thermal: 1.1 W worst case models about 81 C rise against 40 C declared. The BQ24074 folds charge current back at 125 C, so the cost is slower charging when hot.
- U1 ground vias: 4 fit under the pad, the check wants 5.
- +SYS pour neck and 2-via transition near R6: the DONE-LED branch, a few mA.
- GND F.Cu neck beside USB-A J4: the return runs through its pads into the B.Cu plane; add stitching vias if F.Cu continuity is wanted.
- Earlier provisional P2 choices: no input fuse or TVS, no reverse-battery FET, no 5 V limiter, 1 A default charge current.

Route style: 108 segments, 0 arcs, 0 off-angle, two small-jog warnings (a 0.6 mm sidestep on +5V and twelve sidesteps of 0.05 to 0.11 mm on LBO). The full design doc has the detail.

Checks and Cost

GATE	STATUS	ATTEMPTS	FAILING
erc	pass	3	0
place	pass	25	0
drc_routed	pass	5	0
verify	pass	4	0
dfm	pass	3	0

The full design doc (lipo-boost-design-full) has every decision, the renders and the run's history.